

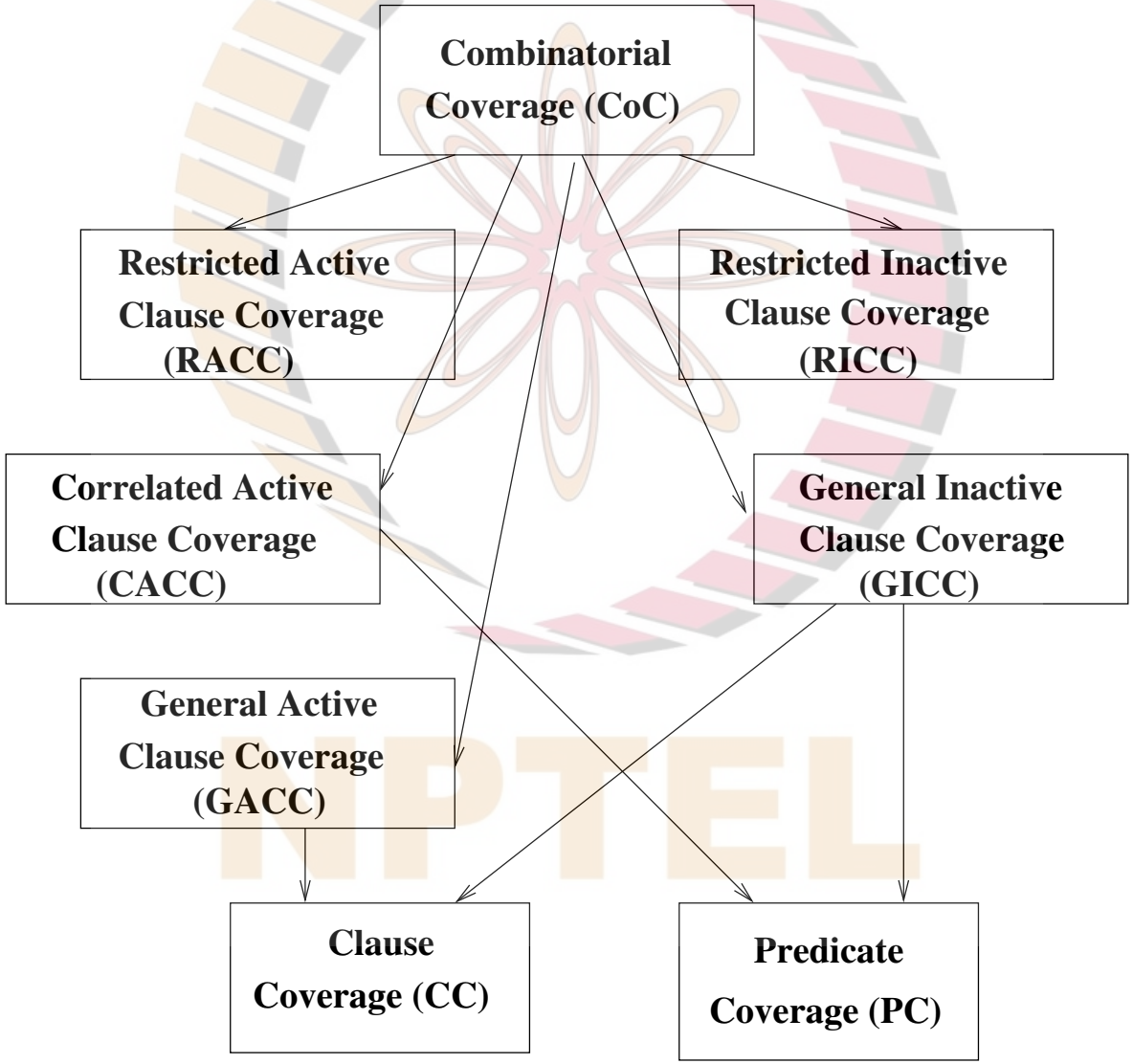
Logic Coverage Criteria: Making clauses determine predicate

Meenakshi D'Souza

International Institute of Information Technology
Bangalore.

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Recap: Logic coverage criteria and their subsumption



Test cases for predicate, clause and combinatorial coverage

- Test cases for satisfying TRs for predicate, clause and combinatorial coverage are easy to define.
- As long as a predicate or a clause is not valid/not a contradiction, we can find test cases to achieve predicate coverage and clause coverage.
- For combinatorial coverage, test cases will correspond to all the values in the truth table.
- The usual problem of reachability (RIPR criteria) will have to be solved for writing test cases.

Test cases for ACC and ICC criteria

- For the various ACC criteria, we have to make the clause chosen as major clause determine a predicate.
- There are several ways of doing this. We discuss one approach here.
- Using the *resulting predicates*, we can write test cases for the various ACC criteria.

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Making a clause determine a predicate

- Consider a predicate p with a clause c.
- Let
  - $p_{c=true}$ represent the predicate p with every occurrence of c replaced with *true* and
  - $p_{c=false}$ represent the predicate p with every occurrence of c replaced with *false*.
- Not that neither $p_{c=true}$ nor $p_{c=false}$ contains any occurrence of c.
- Define $p_c = p_{c=true} \oplus p_{c=false}$, $\oplus$ represents exclusive or.

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Making a clause determine a predicate

- p_c describes the exact conditions under which the value of c determines p.
- Clauses in p_c take values such that p_c is true: c determines p (in terms of truth values).
- Clauses in p_c take values such that p_c is false: p is independent of c (in terms of truth values).

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Making a clause determine a predicate: Examples

- Consider $p = a \vee b$.
- Then,

$$
\begin{aligned}
p_a &= p_{a=true} \oplus p_{a=false} && (1)\\
&= (true \vee b) \oplus (false \vee b) && (2)\\
&= true \oplus b && (3)\\
&= \neg b && (4)
\end{aligned}
$$

- For major clause a to determine p, the only minor clause b must be false.
- Symmetrically, $p_b = \neg a$.

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Making a clause determine a predicate: Examples

- Consider $p = a \wedge b$.
- Then,

$$
\begin{aligned}
p_a &= p_{a=true} \oplus p_{a=false} && (5)\\
&= (true \wedge b) \oplus (false \wedge b) && (6)\\
&= b \oplus false && (7)\\
&= b && (8)
\end{aligned}
$$

- For major clause a to determine p, the only minor clause b must be true.
- Symmetrically, $p_b = a$.

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Making a clause determine a predicate: Examples

- Consider $p = a \equiv b$.
- Then,

$$
\begin{aligned}
p_a &= p_{a=true} \oplus p_{a=false} && (9)\\
&= (true \equiv b) \oplus (false \equiv b) && (10)\\
&= b \oplus \neg b && (11)\\
&= true && (12)
\end{aligned}
$$

- For any value of b, a determines p *without regard to the value for* b.

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ICC and predicates with $\oplus$ and $\equiv$

- For a predicate p where the value of p_c for a clause c turns out to be true, ICC criteria are infeasible with respect to c.
- Typically, ICC criteria are likely to result in infeasible test requirements when applied to predicates with $\equiv$ or $\oplus$ operators.
- Symmetrically, if a predicate p has a clause c such that p_c evaluates to false, ACC criteria are infeasible for p with respect to c.
- Basically, the clause c is redundant; predicate p can be written without it.

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Redundant clause in a predicate: Example

- Consider the predicate $p = a \wedge b \vee a \wedge \neg b$. This is just $p = a$.
- Computing p_b, we get

$$
\begin{aligned}
p_b &= p_{b=true} \oplus p_{b=false} && (13)\\
&= (a \wedge true \vee a \wedge \neg true) \oplus (a \wedge false \vee a \wedge \neg false) && (14)\\
&= (a \vee false) \oplus (false \vee a) && (15)\\
&= a \oplus a && (16)\\
&= false && (17)
\end{aligned}
$$

- It is impossible for b to determine p.

Clauses determining a predicate: More examples

- Consider $p = a \wedge (b \vee c)$.
- Then,

$$
\begin{aligned}
p_a &= p_{a=true} \oplus p_{a=false} && (18)\\
&= (true \wedge (b \vee c) \oplus (false \wedge (b \vee c)) && (19)\\
&= (b \vee c) \oplus false && (20)\\
&= b \vee c && (21)
\end{aligned}
$$

- a determines p when $b \vee c$ is true.
- Three choices make $b \vee c$ true:
  $(b = c = true), (b = true, c = false), (b = false, c = true)$.
- For CACC: Pick one pair when a is true and another when a is false.
- For RACC: Choose the same pair for both values of a.

Clauses determining a predicate: More examples

- Consider $p = a \wedge (b \vee c)$.
- Then,

$$
\begin{aligned}
p_b &= p_{b=true} \oplus p_{b=false} && (22)\\
&= (a \wedge (true \vee c) \oplus (a \wedge (false \vee c)) && (23)\\
&= (a \wedge true) \oplus (a \wedge c) && (24)\\
&= a \oplus (a \wedge c) && (25)\\
&= a \vee \neg c && (26)
\end{aligned}
$$

- Symmetrically, p_c is $a \vee \neg b$.

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TRs for ACC criteria

- ACC criteria can get complicated for certain predicates in terms of finding TRs and subsequently, satisfying test cases.
- We walk through an example illustrating how to find final set of TRs and then, test cases for ACC criteria.

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TRs for ACC criteria: Example

- Consider the predicate $p = (a \vee b) \wedge c$.
- We first give the truth table for p.

	a	b	c	$(a \vee b) \wedge c$
1	T	T	T	T
2	T	T	F	F
3	T	F	T	T
4	T	F	F	F
5	F	T	T	T
6	F	T	F	F
7	F	F	T	F
8	F	F	F	F

- Clauses for p_a, p_b and p_c are given below.

p_a	$\neg b \wedge c$
p_b	$\neg a \wedge c$
p_c	$a \vee b$

TRs for GACC criteria: Example

- TR for GACC: Each major clause be true and false, minor clauses be such that major clause determines the predicate.
- TR for GACC: $\{(a = true \wedge p_a, a = false \wedge p_a), (b = true \wedge p_b, b = false \wedge p_b), (c = true \wedge p_c, c = false \wedge p_c)\}$.
- The following table gives true/false values for TR for GACC:

	a	b	c	p
$a = true \wedge p_a$	**T**	F	T	T
$a = false \wedge p_a$	**F**	F	T	F
$b = true \wedge p_b$	F	**T**	T	T
$b = false \wedge p_b$	F	**F**	T	F
$c = true \wedge p_c$	T	F	**T**	T
$c = false \wedge p_c$	F	T	**F**	F

- First and fifth rows are identical, second and fourth rows are identical. Only four tests are needed to satisfy GACC.

TRs for other ACC: Example

- Similarly, we can compute TR for the example for CACC and RACC criteria.
- Exercise: Compute CACC, show that CACC is the same as GACC.
- CACC is considered good for practical use.

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Credits

Part of the material used in these slides is derived from the presentations of the book Introduction to Software Testing, by Paul Ammann and Jeff Offutt.

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